

1. This system does not have a unique solution. Solve the system if possible.

$$\begin{cases} x - y + 2z = -2 \\ 3x + 4y + 5z = -7 \\ x + 6y + z = -3 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. There are infinitely many solutions. If z is allowed to be any real number, then $x = \boxed{}$ and $y = \boxed{}$.
(Type expressions using z as the variable. Use integers or fractions for any numbers in the expression.)
- ☐ B. There is no solution.

2. This system does not have a unique solution. Solve the system, if possible.

$$\begin{aligned} 13x - 14y - 19z &= 12 \\ 12x - 14y - 18z &= 19 \\ x - z &= 0 \end{aligned}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. There are infinitely many solutions. If z is allowed to be any real number, then $x = \boxed{}$ and $y = \boxed{}$.
(Type expressions using z as the variable.)
- ☐ B. There is no solution.

- *3. Solve the system.

$$\begin{aligned} -5x - 3y + 4z - 6 &= 0 \\ 3x + 6y + 2z - 5 &= 0 \\ -2x - 2y + 6z - 16 &= 0 \end{aligned}$$

The solution set is $\{(\boxed{}, \boxed{}, \boxed{})\}$.
(Type an integer or a simplified fraction.)

- *4. Solve the following system of equations using matrices (row operations). If the system has no solution, say that it is inconsistent.

$$\begin{cases} x - 4y + z = 1 \\ 4x + 2y + 4z = 4 \\ 5x - 2y + 5z = 5 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer box(es) in your choice.

- ☐ A. The solution is $(\boxed{}, \boxed{}, \boxed{})$.
(Simplify your answers.)
- ☐ B. There are infinitely many solutions. The solution can be written as $\{(x, y, z) | x = \boxed{}, y = \boxed{}, z \text{ is any real number}\}$.
(Simplify your answers. Type expressions using z as the variable.)
- ☐ C. There are infinitely many solutions. The solution can be written as $\{(x, y, z) | x = \boxed{}, y \text{ is any real number}, z \text{ is any real number}\}$.
(Simplify your answer. Type an expression using y and z as the variables.)
- ☐ D. The system is inconsistent.

*5. Solve the following system.

$$\begin{cases} \frac{5}{4}x - \frac{4}{3}y + \frac{5}{4}z = -8 \\ \frac{5}{16}x + \frac{3}{4}y - \frac{3}{16}z = -6 \\ \frac{3}{4}x - y + \frac{3}{4}z = -6 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

- ☐ A. The system has a single solution. The solution set is $\{(\text{ } , \text{ } , \text{ })\}$.
(Simplify your answer.)
- ☐ B. There are infinitely many solutions and the equations are dependent. The solution set is $\{(x,y,z) | \text{ } \}$.
- ☐ C. The system is inconsistent. The solution set is the empty set.